In [4]:
```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
from numpy.linalg import svd
```

In [6]:
```
img = mpimg.imread('CAT Img.png')
gray = np.mean(img,axis =2)   #### AXIS = 2 means for img , 0=rows , 1 = colum
U,S,Vt = svd(gray,full_matrices=False)
img_recon = np.stack([R_recon, G_recon, B_recon], axis=2)
```

In [37]:
```
gray = np.mean(img,axis =-1)
```

In [43]:
```
U,S,Vt = svd(gray,full_matrices=False)
```

In [13]:
```
u = U[:,:50]
```

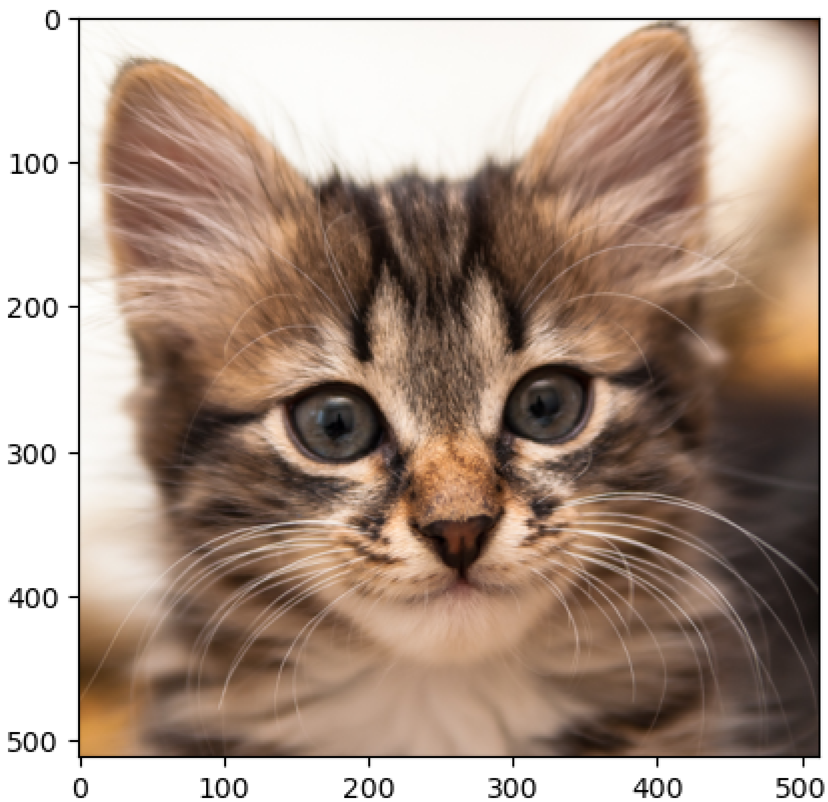
In [14]:
```
v= Vt[:50,:]
```

In [15]:
```
s = np.diag(S[:50])
```

In [10]:
```
red = U[:,:50] @ np.diag(S[:50]) @ Vt[:50,:]
```

In [7]:
```
print(plt.imshow(img))
plt.show()
```

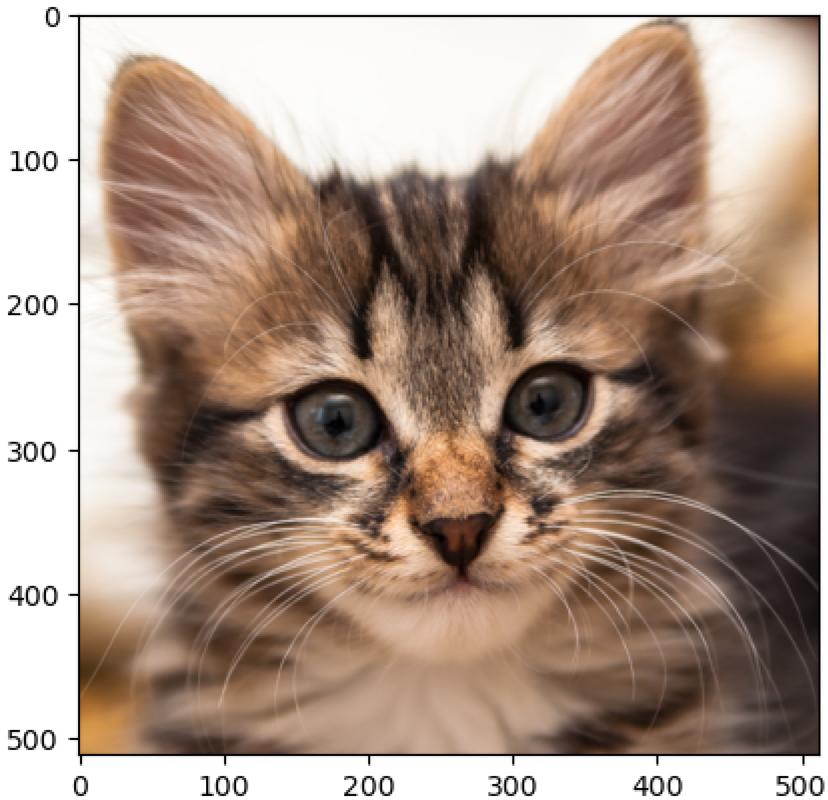
```
AxesImage(shape=(512, 512, 3))
```



In [27]:
```
red1 = U[:,:10] @ np.diag(S[:10]) @ Vt[:10,:]
```

In [8]:
```
print(plt.imshow(img))
```

```
AxesImage(shape=(512, 512, 3))
```

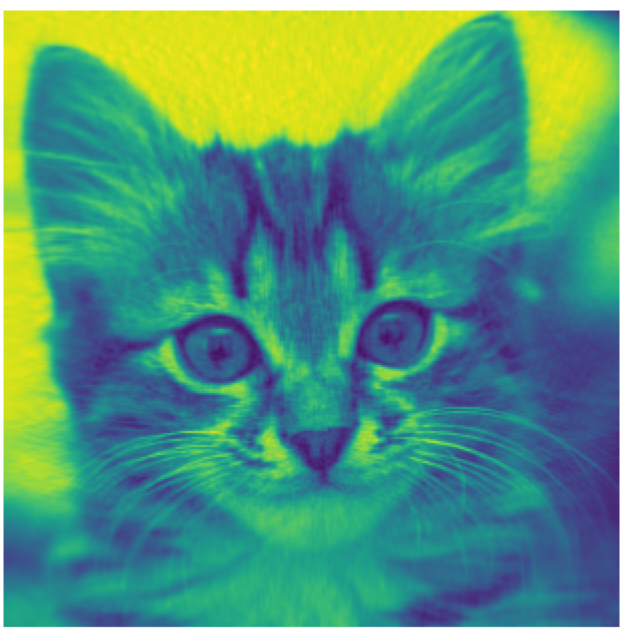



In [29]:
```
red2 = U[:,:] @ np.diag(S[:]) @ Vt[:,:]
```

In [49]:
```
plt.figure(figsize=(4, 4))
plt.axis('off')
print(plt.imshow(img))
plt.show()
```

```
AxesImage(shape=(512, 512, 3))
```

In [11]:
```
plt.figure(figsize=(4, 4))
plt.imshow(red)
plt.axis('off')
plt.show()
#Plot the white and black image version
plt.figure(figsize=(4, 4))
plt.imshow(img,cmap='gray')
plt.axis('off')
plt.show()
```



In [50]:
```
plt.figure(figsize=(4, 4))
plt.imshow(red,cmap='gray')
plt.axis('off')
plt.show()
```

In [15]:
```
red_recon = U[:,:50] @ np.diag(S[:50]) @ Vt[:50,:]
plt.imshow(red_recon , cmap='gray')  # show as red channel
plt.axis('off')
plt.show()
```

```
In [ ]:
```